Disparity analyses are robust to ancestral state estimation uncertainty

Caleb Scutt[1,2*], Natalie Cooper[2], Gavin Thomas[1], Thomas Guillerme[1]

[1]School of Biosciences, University of Sheffield, Sheffield, UK
[2]Science Group, Natural History Museum, London, UK

*Corresponding author: cnscutt1@sheffield.ac.uk

Abstract

Ancestral state estimation is widely used to infer unobserved traits across a phylogeny, extending analyses beyond incomplete fossil records. In studies of morphological disparity ancestral state estimation can account for ghost lineages, yet estimates are often confounded with high uncertainty How best to deal with this uncertainty remains unclear. Using simulations, we compared pre- and post-ordination estimation methods, and both point estimate and probabilistic approaches for treating uncertainty, under continuous and discrete trait models and varying fossil sampling levels. The relative disparity error was calculated from the difference between our estimated disparity values and the true disparity values from the original simulations. We then extended this framework to evaluate the recovery of changes in disparity across time by simulating mass extinction events with varying intensity and selectivity. We find that pre-ordination probabilistic estimations outperform post-ordination and point estimation methods and that omission of ancestral state estimations leads to underestimates of disparity. As expected, higher fossil sampling improves estimation accuracy, although recovered disparity accuracy was robust even at low fossil sampling. Similar patterns were observed when recovering changes in disparity across mass extinction events but declined significantly when fossil sampling was low, highlighting that fossil inclusion is essential when assessing disparity in deep time. These results demonstrate that explicitly propagating uncertainty, rather than collapsing states into point estimates, is more likely to recover accurate patterns in disparity, reinforcing the need to move beyond point estimates when working with the incomplete fossil record.

Keywords: ancestral state estimation; disparity; fossils; simulation; uncertainty; macroevolution

1 Introduction

Morphological diversity, hereafter termed disparity, is widely used for investigating macroevolutionary dynamics (Guillerme et al. 2020a). Disparity can be described as the distribution or spread of taxa within a multidimensional trait space defined by morphological similarity (Erwin 2007) and can be quantified using both continuous (García-Navas et al. 2017) and discrete characters (Foote 1994; Lloyd 2016). Morphological trait datasets are routinely integrated with phylogenies when examining disparity across clades or time. These analyses can be used to investigate selectivity across mass extinctions (Bapst et al. 2012; Cole and Hopkins 2021; Liu et al. 2024), can identify timings of key morphological innovations (Foth et al. 2021). However, such analyses are inevitably limited by the incompleteness of the fossil record; only a biased subset of past diversity is preserved, leaving unsampled ghost lineages and obscuring large portions of a clade's evolutionary history (Flannery Sutherland et al. 2019; Smith et al. 2021). This missing data has been shown to result in both under- and over-estimations of disparity (Mitchell 2015; Smith et al. 2023). For example, high amounts of missing data results in 'centroid-slippage', where an incomplete taxon is pulled towards the centre of the trait space (Flannery Sutherland et al. 2019). Conversely, exceptional preservation, such as Lagerstätten taxa may result in inflated disparity, seen in ichthyosaurs (Flannery Sutherland et al. 2019) and pterosaurs (Butler et al. 2012).

One way in which unsampled lineages can be accounted for is through the use of ancestral state estimation (ASE). Ancestral state estimation is the broad method of estimating trait value(s) in a set of common ancestors given a set of observed data for tips on a phylogenetic tree (Schluter et al. 1997; Cunningham et al. 1998; Revell 2025). ASE has been extensively used across macroevolutionary studies for inferring the likely character state at each node on a phylogeny for continuous or discrete traits of interest. These estimated states can be combined with observed data to generate a more complete trait space, mitigating the gaps left by an incomplete fossil record and, in theory, improve our ability to recover the true disparity signal (Sidlauskas 2008; Brusatte et al. 2011; Lloyd 2018). Inferring ancestral states also allows disparity across time analyses to be explored with higher resolution (Brusatte et al. 2011; Halliday and Goswami 2016; Guillerme 2018).

1.1 Issues with ASE

Despite its theoretical advantages, integration of ancestral state estimation within the disparity analysis pipeline has often been neglected. Much of the apprehension towards the use of ancestral state estimation has likely stemmed from the notion that estimates are inherently uncertain; root-ward estimates in particular are often associated with large confidence intervals or inconclusive scaled likelihoods (Schluter et al. 1997; Losos 1999; Gascuel and Steel 2014). Despite the encouragement for uncertainty to be carried forward in comparative analyses (Silvestro et al. 2015; Boyko and O'Meara 2024), most computational implementations of ancestral state estimation return point estimates, since there is little guidance on how to correctly treat this uncertainty downstream when constructing the trait space. These point estimates are typically averages from confidence intervals for continuous characters or collapsed states returned from arbitrary thresholds (Smith and Donoghue 2022; Casali et al. 2023). By ignoring uncertainty, point estimates can lead to overconfidence in results and erroneous inferences in downstream analyses, all while obscuring potentially accurate signals (Ackerly et al. 2006; Vanderpoorten and Goffinet 2006).

Another key issue with ancestral state estimation is its sensitivity to model misspecification. Likelihood-based approaches to ancestral state estimation require both a tree topology and a specified model of character evolution. Estimation of continuous characters are typically modeled under Brownian motion (BM), which describes a directionless, stochastic process where the trait divergence between related taxa is proportional to their shared evolutionary history and the rate of evolution (Felsenstein 1985; Schluter et al. 1997; O'Meara et al. 2006). However, deviations from BM in the trait data, such as directional trends or stabilising selection, can result in inaccurate estimations. For example, both Webster and Purvis (2002) and Finarelli and Flynn (2006) were unable to recover accurate ancestral estimates of body size in planktonic foraminifera and mammals, respectively, due to a trend of increase through time. Model-related issues can also arise for discrete traits, which are estimated using an Mk model, requiring a tree topology and specified model of evolution (describing transition rates). Estimation accuracy of discrete states appears prone to model over-parameterisation and sensitive to fast transition rates (Keating 2023). Furthermore discrete ancestral state estimation faces a paradoxical trade-off where a slow rate of evolution allows for an easy estimation of ancestral states, but provides little information for estimating transition rates (Gascuel and Steel 2020).

Ané (2008) demonstrated that ancestral state estimation is statistically inconsistent, meaning

that even if the sample size of extant tips were infinite, estimates are not expected to converge on the true value. The inclusion of fossils can improve estimations, due to the shorter evolutionary path to the root of the tree (Finarelli and Flynn 2006; Finarelli and Goswami 2013; Joy et al. 2016; Goswami and Clavel 2025). Numerous simulation and empirical based studies have shown that ancestral state estimations based on extant-only taxa are extremely biased (Webster and Purvis 2002; Finarelli and Flynn 2006; Albert et al. 2009; Betancur-R et al. 2015; Royer-Carenzi and Didier 2016; Monson et al. 2022; Wisniewski et al. 2022; Keating 2023). Similar biases have been observed in disparity analyses (Mitchell 2015). Despite this, many studies are forced to use extant-only trees for both ancestral state estimation (Jetz and Pyron 2018; Amador and Giannini 2021) and disparity (Harmon et al. 2010; Baker et al. 2015; Smith and Donoghue 2022) due to the lack of reasonable fossil data.

1.2 Previous work

Numerous studies have assessed the raw estimation error of ancestral state estimation approaches under different models of evolution, methods of estimation and fossil sampling level (Elliot and Mooers 2014; Royer-Carenzi and Didier 2016; Reyes et al. 2018; Holland et al. 2020; Keating 2023; Boyko et al. 2025). However, how state estimation accuracy is propagated into recovery of an accurate trait space is less understood, despite the continued use of ancestral state estimation in disparity analyses. Lloyd (2018) demonstrated that carrying out ancestral state estimation post-ordination rather than pre-ordination can distort the distribution of points in a trait space markedly, introducing unwanted phylogenetic signal and low levels of convergence. However, this study was restricted to a single empirical dataset and to discrete traits. Thus, we lack a comprehensive understanding of how estimation approaches, trait data and fossil sampling level interact to affect our ability to recover accurate patterns in trait space.

1.3 Aims

Using a simulation framework, we expand on this work to evaluate the ability of different ancestral state estimation approaches to recover true disparity values across different evolutionary models and fossil sampling regimes. Simulations offer a robust approach for benchmarking methodologies since a ground-truth dataset can be generated, so trait values and the resulting disparity signal are all known *a priori* (Lotterhos et al. 2022). Different methodological choices and estimation scenarios can then be evaluated against the true disparity value. Specifically, we directly compare

pre- versus post-ordination estimation approaches, as well as point-estimate versus probabilistic treatments of uncertainty, thereby examining two axes of methodological variation. We also investigate how estimation error varies under different models of trait evolution and fossil sampling regimes. This integrative framework provides a comprehensive evaluation for understanding when ancestral state estimation improves disparity estimation, when it introduces bias, and how uncertainty should be treated when working with incomplete fossil data.

2 Materials and Methods

2.1 Simulation Framework

2.1.1 Phylogenies

All analyses were carried out using R 4.5.2 (R Core Team 2025). We simulated birth-death trees with the speciation (λ) parameter set to 1.0, and extinction (μ) set to 0.9 (Guillerme 2024). Fixing $\lambda = 1.0$ provides a convenient relative time-scale, since only the ratio of extinction to speciation affects the shape and balance of simulated trees. These parameters correspond to a high turnover regime, consistent with the upper-empirical estimates from diversification studies of animal groups (Scholl and Wiens 2016; Pires et al. 2018; Henao Diaz et al. 2019; Morlon et al. 2024). This level of extinction produces realistic evolutionary histories with substantial extant and extinct diversity, while maintaining a positive net diversification rate ($\lambda - \mu > 0$).

We generated three sets of 100 birth-death trees with these parameters until the simulations reached 50, 100 and 150 living taxa, with maximum tree size of 500, 1000 and 1500 tips, respectively. The randomness of the birth-death model meant some simulated trees contained only extinct taxa, so we re-ran each individual simulation until we obtained the desired number of living taxa (50, 100 or 150). This meant that although the parameters were set to $\lambda = 1$ and $\mu = 0.9$, the effective parameters slightly deviated from these input values (see Supplementary Figure S1). Mean net diversification across all simulated trees was 0.30 ± 0.09.

Trees were pruned to extract the crown clade of the tree for downstream analyses. This ensured that all simulated fossil sampling levels shared an identical root node and tree height, providing a consistent framework for downstream analyses. Any tip included in the analyses was a descendant of the most basal node considered, i.e., the MRCA of extant tips, guaranteeing that all ordinations and disparity values were calculated within a consistent clade, allowing a robust assessment of the interaction between fossil sampling and ancestral state estimation.

2.1.2 Trait simulations

For each phylogeny, we simulated continuous and discrete trait data separately. Continuous trait data were simulated under variations of Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models (table 1). We simulated BM with variance = 0.25. We also simulated BM with a trend (BM_t), where the ancestral trait value increased by an increment of 0.1 multiplied by branch length. For OU models, the data were simulated with strong attraction to the optimum (α) (OU_{st}), weak α (OU_w) and with differing optimum and root states (OU_{sh}). Following Gearty et al. (2024), the α parameter was set based on phylogenetic half life ($\frac{ln(2)}{\alpha}$) to account for differing tree heights across simulations (Cooper et al. 2016). For the strong and weak OU models, α was set as 25% and 75% of the tree height respectively. We simulated 20 independent characters for each model (BM, BM_t, OU_w, OU_{st}, OU_{sh}) across each phylogeny.

Discrete trait evolution was simulated using a discrete-time Markov process, where the probability of state transition along each branch was proportional to the product of the transition probability and branch length. We simulated discrete traits across three rate categories, representing fast (0.9), medium (0.1) and slow (0.01) transition rates (table 1). These rates were assigned by generating equal-rates (ER) transition probability matrices. For each category, 85 binary characters and 15 three-state characters were simulated, following a typical scoring ratio found in empirical data (Guillerme and Cooper 2016; Casali et al. 2023).

2.2 Fossil sampling simulation

To simulate different levels of fossil sampling, extinct tips were either retained ('fossilized') or pruned using a Bernoulli sampling method at five different probabilities: 0%, 5%, 15%, 50% and 100%. This method means each extinct tip had an equal probability of being 'fossilized' at each sample level, irrespective of the total tree size. The corresponding trait matrices were subset to match each downsampled tree.

2.3 Ancestral state estimation (ASE)

We estimated ancestral state on the down-sampled matrices using the `dispRity::multi.ace` (Guillerme 2018), which is a wrapper for `ape::ace` (Paradis and Schliep 2019). We used four combinations of two approaches to ancestral state estimation to evaluate each method's accuracy in recovering the true disparity signal. These were then compared to a null model bypassing ancestral state estimation (Fig. 1, Step 4).

2.3.1 Pre-ordination estimation

Ancestral state estimation of trait data can be carried out at two points either side of the ordination step in the disparity pipeline (Lloyd 2018). Pre-ordination estimation (pre-OASE) involves the estimation of the raw trait values for each node. Post-ordination estimation (post-OASE) calculates the continuous scores across each PC axis of an ordination matrix.

Pre-OASE of continuous characters models evolution under a Brownian motion model, where ancestral state values lie within a multivariate normal distribution that is estimated given the observed tip data and the shared evolutionary history among taxa (Schluter et al. 1997; Rohlf 2001). The width of estimated confidence intervals reflects the uncertainty in continuous ancestral estimates, which typically increases with node depth and decreases with the number of informative descendant lineages.

The pre-OASE method for estimating ancestral discrete characters involves estimating the marginal likelihoods for each discrete state at each node, based on a maximum likelihood derived estimate of the Mk transition matrix and the tree topology given (Pagel 1999; Yang 2006). Given that the transition rates were set *a priori* during simulations, the equal-rates (ER) model was used to avoid model misspecification as a confounding factor on estimation accuracy (Keating 2023).

We derived trait spaces from which to estimate disparity through ordination of these estimations. For discrete characters, distance matrices were computed using the Maximum Observed Rescaled Distance metric (MORD) (Lloyd 2016) and ordinated using principal coordinates analyses (PCoA) (`cmdscale()` function in R). Continuous data were ordinated using principal components analysis (PCA) (`prcomp()` function in R).

2.3.2 Post-ordination estimation

For post-OASE, the same ordination step was applied prior to ancestral state estimation to the down-sampled discrete and continuous character matrices to generate a set of tip-only trait spaces. Ancestral node states were then estimated under a Brownian motion model across all PC axes, as detailed in section 2.3.1.

2.3.3 Uncertainty treatment

For each estimation method, we also compared point estimate and probabilistic approaches. The point estimate approach returns a fixed single value for each estimate. For continuous

characters, the point estimate returned was the median of the ancestral estimate confidence interval. For discrete characters, we used a strict majority rule where the state with the highest scaled likelihood was selected at each node for each character. In scenarios where state likelihoods were equal, one state was chosen at random. We also used a relative majority rule, where any state at a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states was collapsed, otherwise states were left as uncertain (supplementary).

The probabilistic approach differs in that uncertainty in estimates are explicitly accounted for by sampling multiple results rather than a single point estimate. For continuous characters, we sampled each state across 100 matrices, with each estimate value lying within a normal distribution centered on the mean ancestral estimation and spread across the estimated confidence interval. Each discrete character state at each node was sampled 100 times according to their scaled likelihoods (resulting in 100 character estimation matrices). In a scenario where there is uncertainty in a binary character for a specific node, e.g., 0.55 for state 0 and 0.45 for state 1, this distribution approach results in a 55% probability of sampling state 0 in each matrix for the respective node (and 45% chance of sampling state 1). Both procedures were implemented using the `sample = 100` argument in the `dispRity::multi.ace` function (Guillerme 2018).

2.4 Disparity analyses

When assessing overall disparity, we wanted to understand how well ancestral state estimation can recover true patterns in size and density in trait space occupancy (Guillerme et al. 2020b). Size-based metrics describe the spread of points in the trait space (Guillerme et al. 2020b). To explore how well the size of the trait space was recovered, we calculated the sum of quantiles. The sum of quantiles calculates the range on each PC axis for the 95% of the data and then sums these ranges across all axes. This metric has similar properties to the often used sum of ranges metric by describing the total trait space exploited (Foote 1992), but with less sensitivity to outliers and sample size. For density, we used the sum of variances (Ciampaglio et al. 2001; Wills 2001) and the average nearest neighbour distance (Foote 1992). Disparity values were calculated across each ancestral state estimation and the true simulated values so that a relative disparity error could be calculated (Fig. 1 Step 5).

$$\text{Relative disparity error} = \frac{\text{Estimated disparity} - \text{True disparity}}{\text{True disparity}} \tag{1}$$

We calculated disparity across three different treatments of the ordinated matrices. First, we used the untreated ordination output to examine how well ancestral state estimation recovers the true disparity signal across all PC axes. Second, we min-max standardised the PC axes so that the absolute magnitude in PC scores did not influence the disparity metric scores (i.e. in this case, the max-score for PC1 was always standardised to 1). Third, we only used the minimum number of dimensions that were contained across all fossil sampling levels for that specific phylogeny so that each fossil sampling level, i.e., the first 48, 98 and 148 dimensions for the 50, 100 and 150 living taxa simulations, respectively. This was to negate any effect that higher-dimensional trait space estimations were more likely to be closer the true disparity signal.

2.5 Statistical tests

The relative disparity error values were converted to absolute errors to measure the magnitude of deviation from the true disparity, regardless of direction. Due to the zero-inflated and right-skewed nature of the error distribution, we applied a log-transformation ($log(|x| + 0.001)$). Since these errors were now log-errors, essentially the more negative the log error the more accurate (closer to 0) the original relative error is. Given the imbalance in sample sizes between the different methods, i.e., distribution approaches return 100 disparity values for every 1 point estimate, we aggregated the distribution outputs by taking the median prior to analyses. We also performed subsequent analyses where the distribution was retained, but each individual point returned by distributions was weighted 1/100th of the single point estimate.

We investigated the drivers of disparity estimation accuracy with linear-mixed models (LMM) using the `lme4` R package (Bates et al. 2015). The log-transformed relative error was treated as the response variable. Based on our experimental design, we fitted a four-way interaction between the estimation method, model of evolution, fossil sampling level and disparity metric. We also included tree size as a further non-interactive fixed effect. To account for the variation with each simulation run, each tree replicate was included as a random intercept. We carried out post-hoc pairwise comparisons using estimated marginal means, adjusted using Tukey's HSD.

3 Results

3.1 Continuous traits

3.1.1 Methods of estimation

For continuous characters, the method of estimation had a significant effect on disparity accuracy ($F = 358,\ p < .001$) (Figure 2, Table **??**). Notably, using ancestral state estimation methods were found to improve estimates of disparity compared to excluding ancestral states when averaged over all scenarios (Table S2). Post-hoc pairwise comparisons of estimated marginal means (*emmeans*) revealed both pre- and post-ordination distributions of ancestral state estimations returned the lowest log disparity errors. In contrast, point estimates performed worse overall than bypassing ancestral state estimation (Table S2).

3.1.2 Interaction effects

However, this pattern was context-specific. Fossil sampling was the strongest driver of recovery of an accurate disparity signal for continuous traits ($F = 9169,\ p < .001$), as each increase in fossil sampling level resulted in a significant reduction in disparity error. Similarly, the evolutionary model ($F = 2956, p < .001$) and disparity metric ($F = 5028, p < .001$) were also strong drivers of variation in error. Tree size had a minor effect in comparison ($F = 43, p < .001$). The three-way interaction between estimation method, fossil sampling and evolutionary model was significant ($F = 60, p < .001$) and the post-hoc estimated marginal means can be used to identify the optimal method for models at each fossil sampling levels. For Brownian motion (BM) at 100% fossil sampling, the distribution estimations were most accurate. At 50% and below, methods of ancestral state estimation generally performed equally well and significantly better than excluding ancestral states. Under BM + trend, a similar pattern was followed at high fossil sampling levels, yet when fossil sampling drops below 15% methods of ancestral state estimation perform worse than excluding ancestral states. In contrast, under an OU model excluding ancestral states was the best approach across all fossil sampling levels. While the disparity metric had a large effect on the overall estimation accuracy ($F = 5028,\ p < .001$), it was a relatively lower driver of variation regarding its interaction with the estimation method ($F = 7, p < .001$). Therefore, while estimation accuracy differed between metrics, the hierarchy between optimal methods was generally consistent across metrics.

Key takeaways:

- Use **pre/post ord distribution** estimations when trait data is BM (+ trend).
- If trait data has a trend, fossil sampling >5% is essential to return reliable disparity signal.
- When trait data is OU, it is most accurate to exclude ancestral states at all fossil sampling levels.

3.2 Discrete traits

3.2.1 Methods of estimation

For discrete characters, the method of estimation also had a significant effect on disparity accuracy ($F = 10767, p < .001$) (Figure 4). Similar to continuous traits, methods of ancestral state estimation consistently improved disparity estimates compared to excluding ancestral states (Table S3). However, the optimal type of estimation differed. Post-hoc pairwise comparisons of estimated marginal means revealed that pre-ordination point estimates returned the lowest log disparity errors overall. In contrast to continuous traits, distribution methods generally performed worse than pre-ordination point estimates on average, but post-ordination point estimates performed worse than excluding ancestral states (Table S3).

3.2.2 Interaction effects

Similar to continuous traits, fossil sampling ($F = 22191,\ p < .001$), transition rate ($F = 6957, p < .001$) and metric ($F = 26174, p < .001$) were again major drivers of variation in accuracy. As expected, increased fossil sampling and decreased transition rate led to lower disparity errors, and the mean pairwise distance was the easiest metric to recover, whereas the sum of quantiles was the hardest. The three-way interaction between estimation method, fossil sampling and transition rate explained much of the variation in the data ($F = 588, p < .001$). The broad pattern was that at high fossil sampling levels, pre-ordination point estimates were the best method, whereas at lower fossil sampling levels both pre- and post-ordination distributions returned the most accurate disparity values. This pattern was model-dependent, however, as under a slow rate the pre-ordination distribution method was the best performing method across all fossil sampling levels. At fast and medium rates under low fossil sampling ($< 15\%$), the post-ordination distribution method returned the lowest disparity errors. Notably, methods of ancestral state estimation outperformed their exclusion across all combinations of transition rate, fossil sampling level and disparity metric. Again, while the disparity metric was a significant

driver to overall accuracy, it had little effect on the hierarchy in methods set by the transition rate and fossil sampling level, signified by a weak four-way interaction ($F = 10, p < .001$.)

Key takeaways:

- Use **pre-ordination point** estimations when fossil sampling is dense >15% and transition rates are medium/fast
- Use **post-ordination distribution** when fossil sampling is low <15% and transition rates are medium/fast
- If rates are slow, the **pre-ordination distribution** method is best across all fossil sampling levels
- It is encouraged to always use ancestral state estimation , since it improve estimations across all scenarios.

4 Discussion

Our results suggest that, overall, the inclusion of ancestral state can improve our estimations of disparity compared to their exclusion. However, we found this pattern to be context-specific, particularly dependent on fossil sampling and model of evolution. Overall, we can recommend that ancestral state estimations be always be used in estimations of disparity from discrete characters, but recommend caution for continuous characters.

4.1 Pre versus post ord

For continuous characters, we found virtually no discrepancy between the effect of pre- and post-ordination estimation of ancestral states on disparity. Given that continuous traits exist within a Euclidean space, the linear transformations of both PCA and ancestral state estimation under Brownian motion (BM) mean that estimated node states exist within the same relative positions to the observed tip data for both pre- and post-ordination estimation.

For discrete traits, the timing of ancestral state estimation was a key driver in disparity accuracy. Overall, pre-ordination estimation led to more accurate disparity estimations than post-ordination estimation, a finding consistent with Lloyd (2018). Post-ordination estimation is ultimately limited by its geometric assumptions. Since axes of the trait space are treated as continuous traits evolving under BM, the ancestral nodes are inevitably constrained within

the convex hull defined by the observed tip data (Stayton 2015; Lloyd 2018; Gerber 2019). This forces the implicit assumption that our observed trait data accounts for the extremes of morphological diversity for the clade of interest and that past diversity was only a subset of modern diversity. Disparity is therefore underestimated, particularly with range-based metrics such as sum of quantiles (Figure 4). Estimation of discrete ancestral states prior to ordination does not encounter these same limitations. A discrete trait matrix can be said to exist within a hypercube, where each vertex corresponds to a unique combination of traits (Gerber 2019). When the ancestral states are estimated prior to ordination, there is no constraint preventing an ancestor from occupying a vertex that lies peripheral to the observed data. Previously authors have interpreted estimated ancestral nodes that fall outside the observed range (convex hull) as an artificial inflation of disparity (Brusatte et al. 2011). However, as Gerber (2019) clarifies, this interpretation is flawed, since the convex hull is a two-dimensional projection of the true trait space. As exhibited in Figure 4, underestimation of disparity is ubiquitous to all discrete character estimation methods. Therefore, pre-ordination discrete estimations are not bound to a cloud of averages in the way post-ordination estimation is and thus are optimal at recovering true trait space expanse.

4.2 Treatment of uncertainty

The choice of estimation method for continuous characters was largely driven by the treatment of uncertainty. At high fossil sampling levels, incorporating uncertainty through the use of distributions performed better than point estimates (Figure 2; Figure 3). This advantage stems from the tendency of point estimates to smooth over evolutionary history (Boyko and O'Meara 2024; Boyko et al. 2025), underestimating the variance of the true evolutionary path. However, in contrast to discrete traits, continuous estimation distributions are unbounded and thus sensitive to data quality. Therefore, the variation introduced from sampling across a distribution was dependent on the density of fossils; at high fossil density, confidence intervals are constrained due to higher certainty in estimates. As fossil sampling decreases, these confidence intervals are widened and more noise is introduced as a result of using distribution, negating the advantage observed at higher fossil sampling.

For discrete characters, we found an inverse pattern where distribution methods were optimal at low fossil sampling levels and point estimates were optimal at high fossil sampling levels. This is due to the bounded nature of discrete traits; even with minimal information, the disparity

error that can be returned is bounded by the number of states present at each node, which contrasts the effectively infinite error that can be returned by continuous characters. In terms of the pre-ordination estimations, our results reveal two issues relating to uncertainty in estimations. At low fossil sampling or slow transition rates, the predominant issue facing the recovery of a true trait space is the lack of data. For fossil sampling, the lack of evidence in deeper time means our ancestral state estimates, and subsequent trait space pattern, are biased by the living tip data. In the case of slow transition rates, we face a paradoxical situation with slow rates where the root state can be easily estimated, yet the few transitions make the rate of change hard to estimateGascuel and Steel (2020). If we use the hypercube analogy again, the state scaled likelihoods at each node under slow rates or low/absent fossils will be biased against transitions and when point estimates are used ancestral nodes will be constrained within a certain region of vertices in the trait space. In contrast, pre-ordination distribution estimations allow the exploration of other vertices by sampling across alternative discrete state combinations. In the case of slow rates and low fossil sampling, this was found to improve disparity estimations. The opposite side of the spectrum of discrete state 'uncertainty' is the issue of saturated data, where the paradox is flipped and instead, we have an abundance of information, thus transition rates are easy to estimate, yet due to the rate of change most nodes are left with inconclusive scaled likelihoods. Under the fast rates simulated in this study, character transitions are saturated, so there is very little phylogenetic signal. The use of distributions amplifies the noise in this scenario, thus point estimates can be used to filter out the noise.

4.3 Model misspecification & Fossil inclusion

As expected, we found that the differences in disparity errors between models of evolution reflect model misspecification in ancestral state estimation. The inclusion of ancestral state estimations in calculation of disparity under Brownian motion were more accurate than excluding ancestral states across all fossil sampling levels and disparity metrics. This is intuitive, since ancestral state estimation of continuous characters uses the BM model as an assumption for the evolution of traits. The assumptions of BM include a fixed mean and trait variance increasing linearly through time (Felsenstein 1985; O'Meara et al. 2006). We found that when these assumptions of BM were violated, such as under BM + trend and OU models, disparity errors returned by methods of ancestral state estimation increased. For example, under low/absent fossil sampling, methods of ancestral state estimation performed worse than bypassing ancestral state estimation for BM +

trend. In this case, disparity estimates tended to be underestimated, since state estimations will be intermediates of the observed tip data, absent to the directional trend that occurred and thus missing past diversity. This finding supports previous work that highlights the critical impact of fossil inclusion when directional trends are present in the trait data (Finarelli and Flynn 2006; Slater et al. 2012; Finarelli and Goswami 2013). We found that there is a crossover between 5-15% fossil sampling where ancestral state estimation becomes optimal, which would equate to a densely sampled empirical phylogeny (Figure 3). While not tested in this study, we can also assume that disparity metrics that examine changes in position, such as average displacement (Guillerme et al. 2020b), would be vastly underestimated without sufficient fossil sampling.

Similarly, calculations of disparity derived from OU simulations were also underestimated when using ancestral state estimations. The OU process is characterised by a stationary variance, which contrasts the BM assumption of an increase in variance through time. Therefore, since BM estimates internal nodes as a weighted average of the observed tip data, the ancestral nodes are compressed towards the centroid of the trait space. This explains why, even with 100% fossil sampling, bypassing ancestral state estimation always outperformed the use of ancestral state estimation, since the fundamental model of estimation was incorrect. However, in the case of the OU-derived disparity, disparity values returned from ancestral state estimation were not necessarily far off those that bypassed ancestral state estimation, so the decision whether to use ancestral state estimation in this scenario is a trade-off between the slight decreased accuracy of estimations versus the advantages of incorporating ancestral states in disparity analyses.

4.4 Disparity metric & Recovering a true trait space

Previously, the choice of disparity metric has been probed as a critical step in the disparity pipeline, specifically for examining different aspects of trait space (Guillerme et al. 2020b). Indeed, our results do exhibit significant differences in disparity error between different metrics. For example, recovering the sum of quantiles was ultimately harder than recovering the sum of variances or mean pairwise distance when using discrete traits. However, the hierarchy of methods was relatively consistent across disparity metrics, suggesting that the relative performance of ancestral state estimation methods is robust to different disparity metrics. Consistent performance across multiple disparity metrics can reassure us that multiple aspects of the true trait space are recovered, including size, dispersion and density. Where there is inconsistency in methods across metrics, then we can infer that the true disparity value has been attained artificially, rather

than the recovery of a true trait space pattern. This can be observed in the post-ordination distribution estimation of discrete traits, which performed well under the sum of variances and mean pairwise distance metrics, but noticeably worse under sum of quantiles.

4.5 Conclusions

5 Conclusions

Acknowledgments

Author Contributions

Data Availability

Competing Interests

Figures

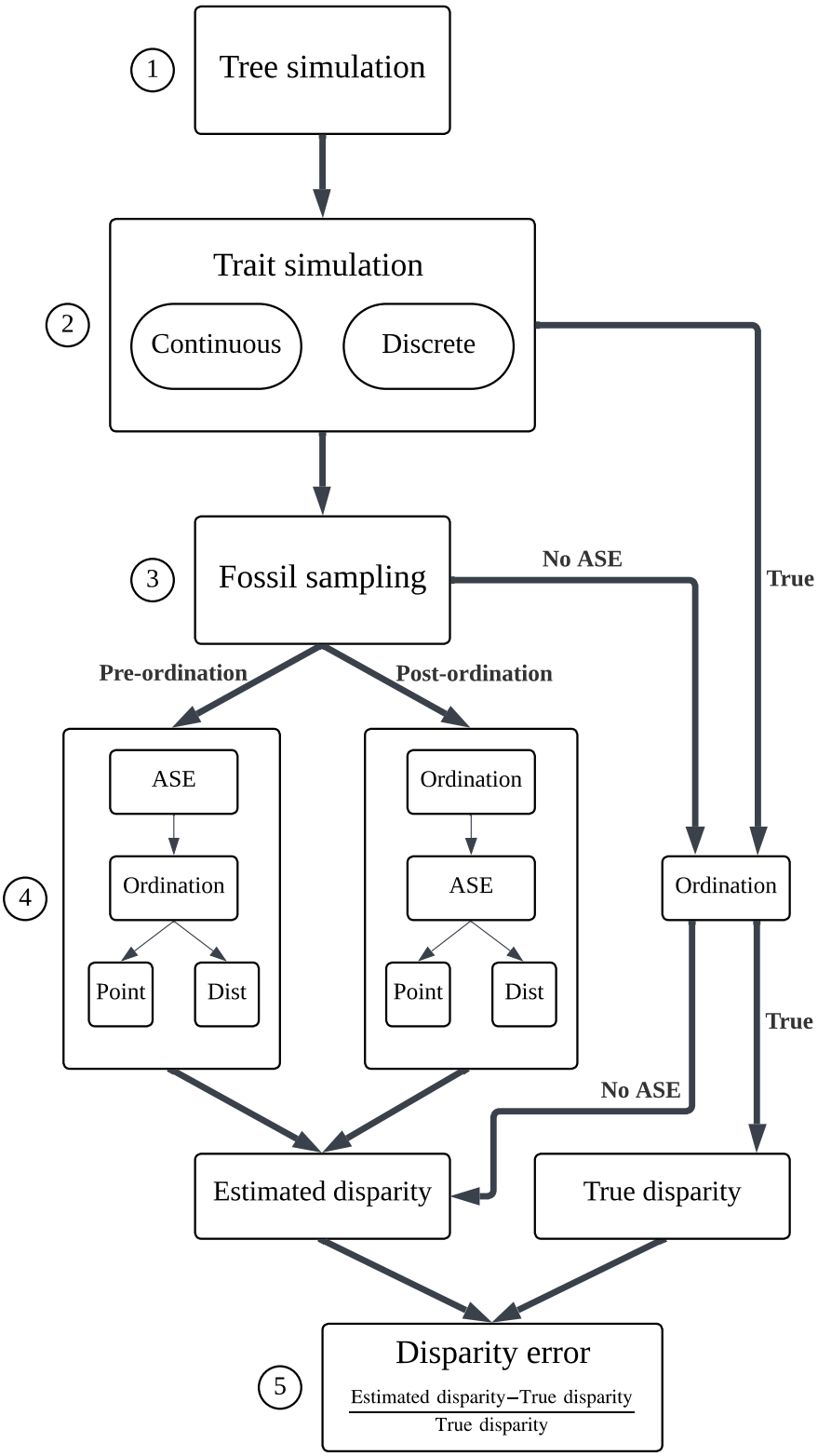

Figure 1: **Simulation workflow**

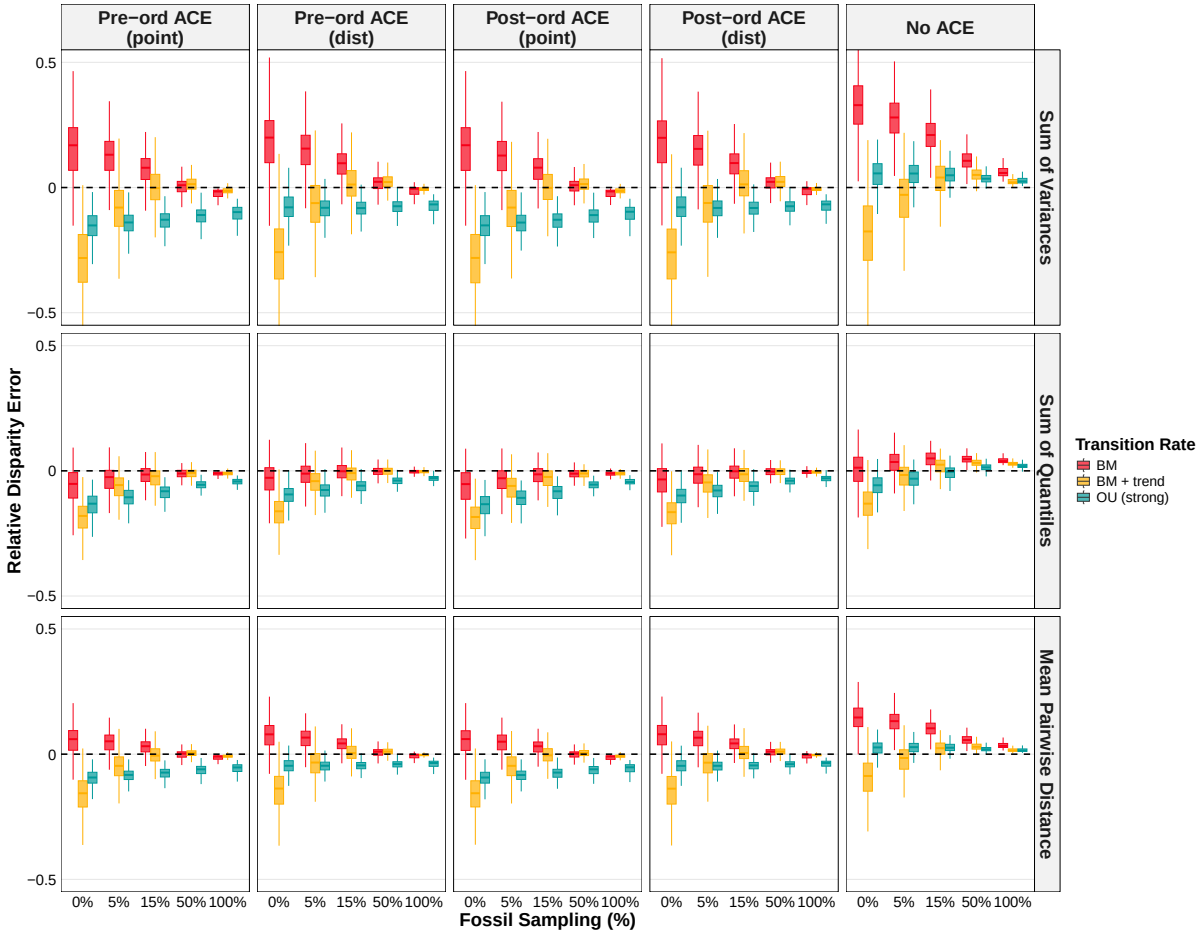

Figure 2: **Results continuous** Continuous

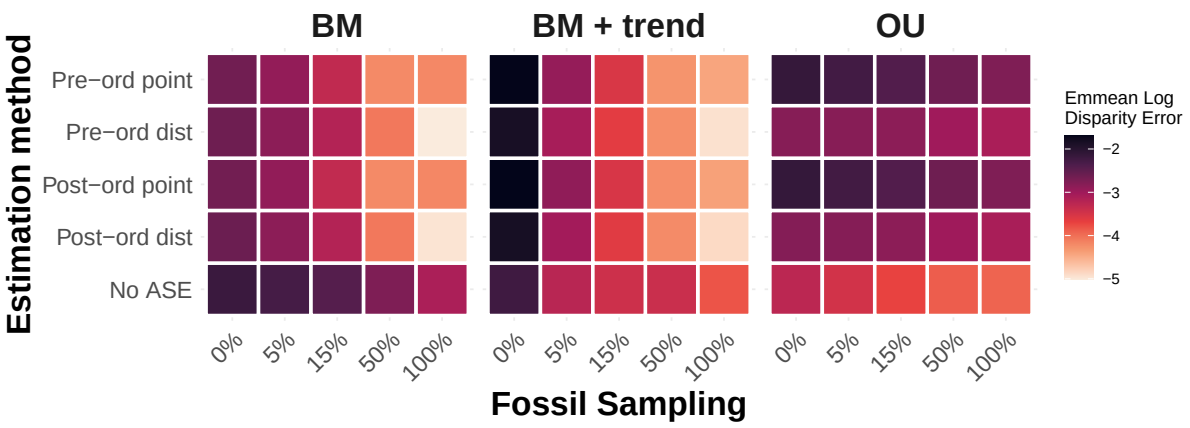

Figure 3: **Heatmap continuous** Continuous

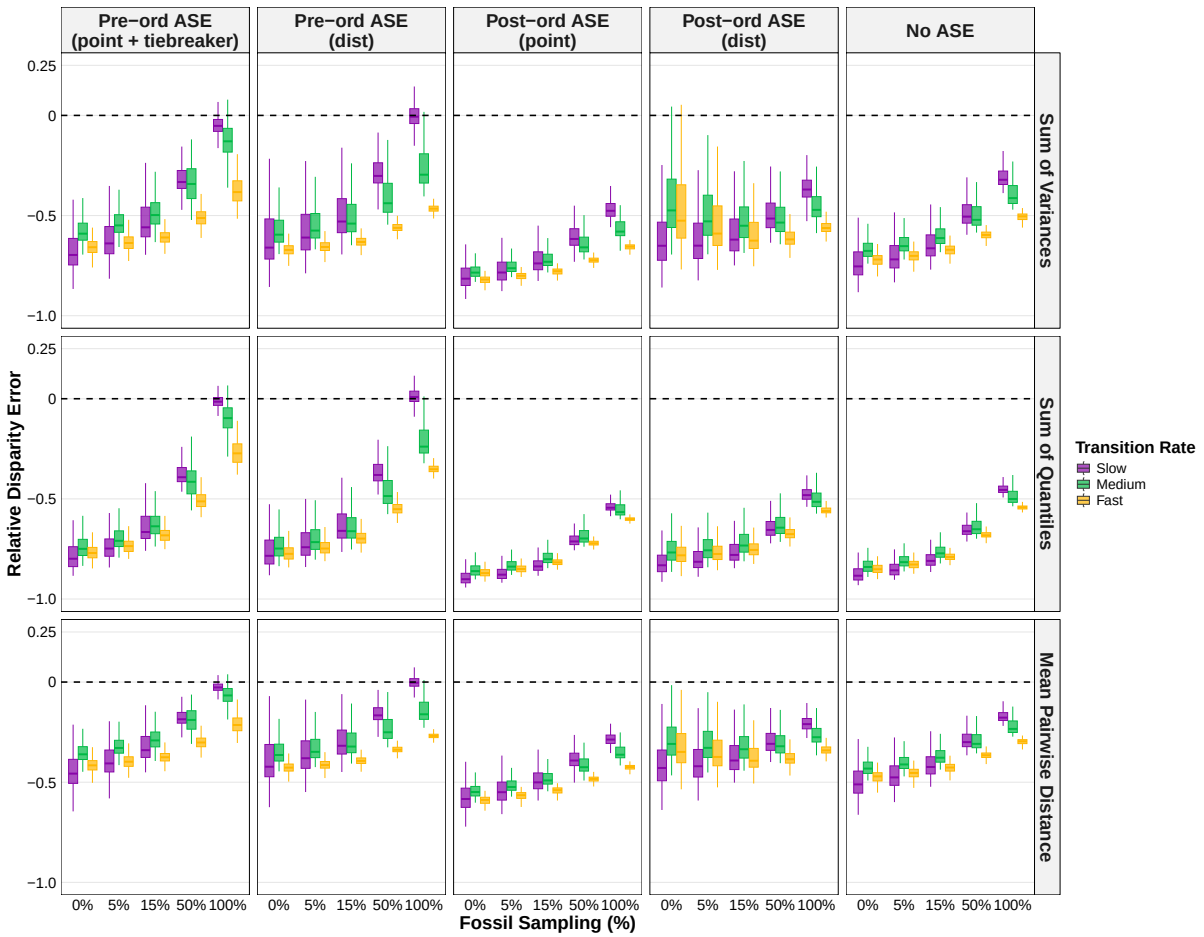

Figure 4: **Results discrete** Relative disparity errors across all dimensions returned by ordinations.

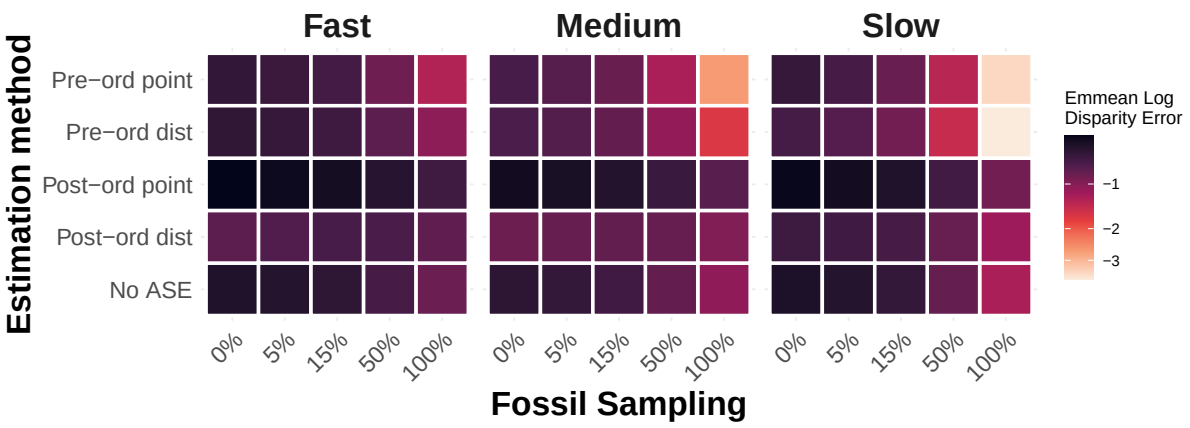

Figure 5: **Heatmap discrete** Continuous

Tables

Table 1: Trait simulation models and parameters used across each phylogeny. For continuous characters, the root state was always set as 0. P_{ij} = transition probability; σ = variance; α = rate of adaptation toward the optimum; θ = optimum.

Type of data	**Model**	**Parameters**
Discrete	$Fast$	$P_{ij} = 0.9$
	$Medium$	$P_{ij} = 0.1$
	$Slow$	$P_{ij} = 0.01$
Continuous	BM	$\sigma = 0.25$
	$BM + trend$	$\sigma = 0.25,\ \text{trend} = 0.1$
	OU_{st}	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.25}$
	OU_{w}	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.75}$
	OU_{sh}	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height} \times 0.75},\ \theta = 2$

Table 2: **Continuous traits: best performing estimation method for each combination of model and fossil sample level.** The best performing method was selected as the most negative estimated marginal mean. In some cases, the method with the lowest log error was not significantly different to other methods, in which case it is denoted with '*'. See

A) Sum of Quantiles

	0%	**5%**	**15%**	**50%**	**100%**
BM	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*
BM + trend	No ASE	No ASE	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*
OU	No ASE	No ASE	No ASE	No ASE	No ASE

B) Sum of Variances

	0%	**5%**	**15%**	**50%**	**100%**
BM	Post-ord point*	Post-ord point*	Post-ord point*	Post-ord point*	Pre-ord dist*
BM + trend	No ASE	No ASE*	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*
OU	No ASE	No ASE	No ASE	No ASE	No ASE

C) Mean Pairwise Distance

	0%	**5%**	**15%**	**50%**	**100%**
BM	Post-ord point*	Post-ord point*	Post-ord point*	Post-ord point*	Pre-ord dist*
BM + trend	No ASE	No ASE*	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*
OU	No ASE	No ASE	No ASE	No ASE	No ASE

Table 3: **Discrete traits: best performing estimation method for each combination of model and fossil sample level.** * means tie (not significantly different to others, but the lowest emmean).

A) Sum of Quantiles

	0%	**5%**	**15%**	**50%**	**100%**
Fast	Post-ord dist*	Pre-ord point*	Pre-ord point*	Pre-ord point	Pre-ord point
Medium	Pre-ord dist*	Pre-ord point*	Pre-ord point*	Pre-ord point	Pre-ord point
Slow	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*	Pre-ord dist*

B) Sum of Variances

	0%	**5%**	**15%**	**50%**	**100%**
Fast	Post-ord dist	Post-ord dist	Post-ord dist	Pre-ord point	Pre-ord point
Medium	Post-ord dist	Post-ord dist	Post-ord dist*	Pre-ord point	Pre-ord point
Slow	Pre-ord dist*	Pre-ord dist	Pre-ord dist	Pre-ord dist	Pre-ord dist

C) Mean Pairwise Distance

	0%	**5%**	**15%**	**50%**	**100%**
Fast	Post-ord dist	Post-ord dist	Post-ord dist*	Pre-ord point	Pre-ord point
Medium	Post-ord dist	Post-ord dist	Pre-ord point*	Pre-ord point	Pre-ord point
Slow	Pre-ord dist*	Pre-ord dist	Pre-ord dist	Pre-ord dist	Pre-ord dist

References

DD Ackerly, DW Schwilk, and CO Webb. Niche evolution and adaptive radiation: testing the order of trait divergence. *Ecology*, 87(sp7):S50–S61, 2006.

James S Albert, Derek M Johnson, and Jason H Knouft. Fossils provide better estimates of ancestral body size than do extant taxa in fishes. *Acta Zoologica*, 90:357–384, 2009.

Lucila I Amador and Norberto P Giannini. Evolution of diet in extant marsupials: emergent patterns from a broad phylogenetic perspective. *Mammal Review*, 51(2):178–192, 2021.

Cécile Ané. Analysis of comparative data with hierarchical autocorrelation. 2008.

Joanna Baker, Andrew Meade, Mark Pagel, and Chris Venditti. Adaptive evolution toward larger size in mammals. *Proceedings of the National Academy of Sciences*, 112(16):5093–5098, 2015.

David W Bapst, Peter C Bullock, Michael J Melchin, H David Sheets, and Charles E Mitchell. Graptoloid diversity and disparity became decoupled during the ordovician mass extinction. *Proceedings of the National Academy of Sciences*, 109(9):3428–3433, 2012.

Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. Fitting linear mixed-effects models using lme4. *Journal of statistical software*, 67:1–48, 2015.

Ricardo Betancur-R, Guillermo Ortí, and Robert Alexander Pyron. Fossil-based comparative analyses reveal ancient marine ancestry erased by extinction in ray-finned fishes. *Ecology letters*, 18(5):441–450, 2015.

James D Boyko and Brian C O'Meara. dentist: Quantifying uncertainty by sampling points around maximum likelihood estimates. *Methods in Ecology and Evolution*, 15(4):628–638, 2024.

James D Boyko, Kyle J Gontjes, Evan S Snitkin, and Stephen A Smith. Uncertainty in joint ancestral state reconstruction: Improving accuracy and biological interpretability of ancestral state prediction. *bioRxiv*, pages 2025–10, 2025.

Stephen L Brusatte, Shaena Montanari, Hong-yu Yi, and Mark A Norell. Phylogenetic corrections for morphological disparity analysis: new methodology and case studies. *Paleobiology*, 37(1): 1–22, 2011.

Richard J Butler, Stephen L Brusatte, Brian Andres, and Roger BJ Benson. How do geological sampling biases affect studies of morphological evolution in deep time? a case study of pterosaur (reptilia: Archosauria) disparity. *Evolution*, 66(1):147–162, 2012.

Daniel M Casali, Alberto Boscaini, Timothy J Gaudin, and Fernando A Perini. Morphological disparity and evolutionary rates of cranial and postcranial characters in sloths (mammalia, pilosa, folivora). *Palaeontology*, 66(1):e12639, 2023.

Charles N Ciampaglio, Matthieu Kemp, and Daniel W McShea. Detecting changes in morphospace occupation patterns in the fossil record: characterization and analysis of measures of disparity. *Paleobiology*, 27(4):695–715, 2001.

Selina R Cole and Melanie J Hopkins. Selectivity and the effect of mass extinctions on disparity and functional ecology. *Science Advances*, 7(19):eabf4072, 2021.

Natalie Cooper, Gavin H Thomas, Chris Venditti, Andrew Meade, and Rob P Freckleton. A cautionary note on the use of ornstein uhlenbeck models in macroevolutionary studies. *Biological journal of the Linnean Society*, 118(1):64–77, 2016.

Clifford W Cunningham, Kevin E Omland, and Todd H Oakley. Reconstructing ancestral character states: a critical reappraisal. *Trends in ecology & evolution*, 13(9):361–366, 1998.

Michael G Elliot and Arne Ø Mooers. Inferring ancestral states without assuming neutrality or gradualism using a stable model of continuous character evolution. *BMC evolutionary biology*, 14(1):226, 2014.

Douglas H. Erwin. DISPARITY: MORPHOLOGICAL PATTERN AND DEVELOPMENTAL CONTEXT. *Palaeontology*, 50(1):57–73, January 2007. ISSN 1475-4983. doi: 10.1111/j.1475-4983.2006.00614.x. Publisher: John Wiley & Sons, Ltd.

Joseph Felsenstein. Phylogenies and the comparative method. *The American Naturalist*, 125(1):1–15, 1985.

John A Finarelli and John J Flynn. Ancestral state reconstruction of body size in the caniformia (carnivora, mammalia): the effects of incorporating data from the fossil record. *Systematic Biology*, 55(2):301–313, 2006.

John A Finarelli and Anjali Goswami. Potential pitfalls of reconstructing deep time evolutionary history with only extant data, a case study using the canidae (mammalia, carnivora). *Evolution*, 67(12):3678–3685, 2013.

Joseph T Flannery Sutherland, Benjamin C Moon, Thomas L Stubbs, and Michael J Benton. Does exceptional preservation distort our view of disparity in the fossil record? *Proceedings of the Royal Society B*, 286(1897):20190091, 2019.

Mike Foote. Rarefaction analysis of morphological and taxonomic diversity. *Paleobiology*, 18(1): 1–16, 1992.

Mike Foote. Morphological disparity in ordovician-devonian crinoids and the early saturation of morphological space. *Paleobiology*, 20(3):320–344, 1994.

Christian Foth, Roland B Sookias, and Martín D Ezcurra. Rapid initial morphospace expansion and delayed morphological disparity peak in the first 100 million years of the archosauromorph evolutionary radiation. *Frontiers in Earth Science*, 9:723973, 2021.

Vicente García-Navas, Víctor Noguerales, Pedro J Cordero, and Joaquín Ortego. Phenotypic disparity in iberian short-horned grasshoppers (acrididae): the role of ecology and phylogeny. *BMC Evolutionary Biology*, 17(1):109, 2017.

Olivier Gascuel and Mike Steel. Predicting the ancestral character changes in a tree is typically easier than predicting the root state. *Systematic biology*, 63(3):421–435, 2014.

Olivier Gascuel and Mike Steel. A darwinian uncertainty principle. *Systematic Biology*, 69(3): 521–529, 2020.

William Gearty, Bethany Allen, Pedro L Godoy, and Alfio Alessandro Chiarenza. The impact of tip age distribution on reconstructing trait evolution using phylogenetic comparative methods. 2024.

Sylvain Gerber. Use and misuse of discrete character data for morphospace and disparity analyses. *Palaeontology*, 62(2):305–319, 2019.

Anjali Goswami and Julien Clavel. Morphological evolution in a time of phenomics. *Paleobiology*, 51(1):195–213, 2025.

Thomas Guillerme. disprity: a modular r package for measuring disparity. *Methods in Ecology and Evolution*, 9(7):1755–1763, 2018.

Thomas Guillerme. treats: A modular r package for simulating trees and traits. *Methods in Ecology and Evolution*, 15(4):647–656, 2024.

Thomas Guillerme and Natalie Cooper. Effects of missing data on topological inference using a total evidence approach. *Molecular phylogenetics and evolution*, 94:146–158, 2016.

Thomas Guillerme, Natalie Cooper, Stephen L Brusatte, Katie E Davis, Andrew L Jackson, Sylvain Gerber, Anjali Goswami, Kevin Healy, Melanie J Hopkins, Marc EH Jones, et al. Disparities in the analysis of morphological disparity. *Biology letters*, 16(7):20200199, 2020a.

Thomas Guillerme, Mark N Puttick, Ariel E Marcy, and Vera Weisbecker. Shifting spaces: Which disparity or dissimilarity measurement best summarize occupancy in multidimensional spaces? *Ecology and evolution*, 10(14):7261–7275, 2020b.

Thomas John Dixon Halliday and Anjali Goswami. Eutherian morphological disparity across the end-cretaceous mass extinction. *Biological Journal of the Linnean Society*, 118(1):152–168, 2016.

Luke J Harmon, Jonathan B Losos, T Jonathan Davies, Rosemary G Gillespie, John L Gittleman, W Bryan Jennings, Kenneth H Kozak, Mark A McPeek, Franck Moreno-Roark, Thomas J Near, et al. Early bursts of body size and shape evolution are rare in comparative data. *Evolution*, 64(8):2385–2396, 2010.

L Francisco Henao Diaz, Luke J Harmon, Mauro TC Sugawara, Eliot T Miller, and Matthew W Pennell. Macroevolutionary diversification rates show time dependency. *Proceedings of the National Academy of Sciences*, 116(15):7403–7408, 2019.

Barbara R Holland, Saan Ketelaar-Jones, Aidan R o'Mara, Michael D Woodhams, and Gregory J Jordan. Accuracy of ancestral state reconstruction for non-neutral traits. *Scientific Reports*, 10(1):7644, 2020.

Walter Jetz and R Alexander Pyron. The interplay of past diversification and evolutionary isolation with present imperilment across the amphibian tree of life. *Nature ecology & evolution*, 2(5):850–858, 2018.

Jeffrey B Joy, Richard H Liang, Rosemary M McCloskey, T Nguyen, and Art FY Poon. Ancestral reconstruction. *PLoS computational biology*, 12(7):e1004763, 2016.

Joseph N Keating. What is the best method for estimating ancestral states from discrete characters? *bioRxiv*, pages 2023–08, 2023.

Xiaokang Liu, Haijun Song, Daoliang Chu, Xu Dai, Fengyu Wang, and Daniele Silvestro. Heterogeneous selectivity and morphological evolution of marine clades during the permian–triassic mass extinction. *Nature Ecology & Evolution*, 8(7):1248–1258, 2024.

Graeme T Lloyd. Estimating morphological diversity and tempo with discrete character-taxon matrices: implementation, challenges, progress, and future directions. *Biological Journal of the Linnean Society*, 118(1):131–151, 2016.

Graeme T Lloyd. Journeys through discrete-character morphospace: synthesizing phylogeny, tempo, and disparity. *Palaeontology*, 61(5):637–645, 2018.

Jonathan B Losos. Uncertainty in the reconstruction of ancestral character states and limitations on the use of phylogenetic comparative methods, 1999.

Katie E Lotterhos, Matthew C Fitzpatrick, and Heath Blackmon. Simulation tests of methods in evolution, ecology, and systematics: pitfalls, progress, and principles. *Annual Review of Ecology, Evolution, and Systematics*, 53(1):113–136, 2022.

Jonathan S Mitchell. Extant-only comparative methods fail to recover the disparity preserved in the bird fossil record. *Evolution*, 69(9):2414–2424, 2015.

Tesla A Monson, Marianne F Brasil, Michael C Mahaney, Christopher A Schmitt, Catherine E Taylor, and Leslea J Hlusko. Keeping 21st century paleontology grounded: Quantitative genetic analyses and ancestral state reconstruction re-emphasize the essentiality of fossils. *Biology*, 11 (8):1218, 2022.

Hélène Morlon, Jérémy Andréoletti, Joëlle Barido-Sottani, Sophia Lambert, Benoît Perez-Lamarque, Ignacio Quintero, Viktor Senderov, and Pierre Veron. Phylogenetic insights into diversification. *Annual Review of Ecology, Evolution, and Systematics*, 55, 2024.

Brian C O'Meara, Cécile Ané, Michael J Sanderson, and Peter C Wainwright. Testing for different rates of continuous trait evolution using likelihood. *Evolution*, 60(5):922–933, 2006.

Mark Pagel. The maximum likelihood approach to reconstructing ancestral character states of discrete characters on phylogenies. *Systematic biology*, 48(3):612–622, 1999.

Emmanuel Paradis and Klaus Schliep. ape 5.0: an environment for modern phylogenetics and evolutionary analyses in r. *Bioinformatics*, 35(3):526–528, 2019.

Mathias M Pires, Brian D Rankin, Daniele Silvestro, and Tiago B Quental. Diversification dynamics of mammalian clades during the k–pg mass extinction. *Biology letters*, 14(9):20180458, 2018.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2025. URL `https://www.R-project.org/`.

Liam J Revell. Ancestral state reconstruction of phenotypic characters. *Evolutionary Biology*, pages 1–25, 2025.

Elisabeth Reyes, Sophie Nadot, Maria Von Balthazar, Jürg Schönenberger, and Hervé Sauquet. Testing the impact of morphological rate heterogeneity on ancestral state reconstruction of five floral traits in angiosperms. *Scientific Reports*, 8(1):9473, 2018.

F James Rohlf. Comparative methods for the analysis of continuous variables: geometric interpretations. *Evolution*, 55(11):2143–2160, 2001.

Manuela Royer-Carenzi and Gilles Didier. A comparison of ancestral state reconstruction methods for quantitative characters. *Journal of Theoretical Biology*, 404:126–142, 2016.

Dolph Schluter, Trevor Price, Arne Ø Mooers, and Donald Ludwig. Likelihood of ancestor states in adaptive radiation. *Evolution*, 51(6):1699–1711, 1997.

Joshua P Scholl and John J Wiens. Diversification rates and species richness across the tree of life. *Proceedings of the Royal Society B: Biological Sciences*, 283(1838):20161334, 2016.

Brian Sidlauskas. Continuous and arrested morphological diversification in sister clades of characiform fishes: a phylomorphospace approach. *Evolution*, 62(12):3135–3156, 2008.

Daniele Silvestro, Anna Kostikova, Glenn Litsios, Peter B Pearman, and Nicolas Salamin. Measurement errors should always be incorporated in phylogenetic comparative analysis. *Methods in Ecology and Evolution*, 6(3):340–346, 2015.

Graham J Slater, Luke J Harmon, and Michael E Alfaro. Integrating fossils with molecular phylogenies improves inference of trait evolution. *Evolution*, 66(12):3931–3944, 2012.

Thomas J Smith and Philip CJ Donoghue. Evolution of fungal phenotypic disparity. *Nature Ecology & Evolution*, 6(10):1489–1500, 2022.

Thomas J Smith, Mark N Puttick, Joseph E O'Reilly, Davide Pisani, and Philip CJ Donoghue. Phylogenetic sampling affects evolutionary patterns of morphological disparity. *Palaeontology*, 64(6):765–787, 2021.

Thomas J Smith, Robert S Sansom, Davide Pisani, and Philip CJ Donoghue. Fossilization can mislead analyses of phenotypic disparity. *Proceedings of the Royal Society B*, 290(2004): 20230522, 2023.

C Tristan Stayton. The definition, recognition, and interpretation of convergent evolution, and two new measures for quantifying and assessing the significance of convergence. *Evolution*, 69 (8):2140–2153, 2015.

Alain Vanderpoorten and B Goffinet. Mapping uncertainty and phylogenetic uncertainty in ancestral character state reconstruction: an example in the moss genus brachytheciastrum. *Systematic Biology*, 55(6):957–971, 2006.

Andrea J Webster and Andy Purvis. Testing the accuracy of methods for reconstructing ancestral states of continuous characters. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, 269(1487):143–149, 2002.

Matthew A Wills. Morphological disparity: a primer. pages 55–144, 2001.

Anna L Wisniewski, Graeme T Lloyd, and Graham J Slater. Extant species fail to estimate ancestral geographical ranges at older nodes in primate phylogeny. *Proceedings of the Royal Society B*, 289(1975):20212535, 2022.

Ziheng Yang. *Computational molecular evolution.* OUP Oxford, 2006.